Subarachnoid haemorrhage: grading and early care

Grading

- **WFNS** (GCS ± motor deficit, I–V); Hunt & Hess I–V
- Modified Fisher (CT blood thickness + IVH, 0–4): predicts vasospasm

Before securing the aneurysm

- SBP < 160 (labetalol, nicardipine); analgesia, antiemetics
- **Nimodipine 60 mg 4-hourly for 21 days**; EVD for hydrocephalus
- Secure early (< 24–72 h): coiling preferred (ISAT) or clipping

Complications

- Neuro: rebleeding, hydrocephalus, DCI, seizures
- Systemic: neurogenic pulmonary oedema, Takotsubo, arrhythmia, hyponatraemia

Key point: Secure the aneurysm early: rebleeding is the first killer.

Notes: SAH p.4

Delayed cerebral ischaemia and hyponatraemia

DCI

- New focal deficit or ↓ GCS ≥ 2 for ≥ 1 h (days 4–14), not explained otherwise
- Vasospasm (TCD velocities > 120 cm/s, Lindegaard > 3) is one cause; microthrombi, CSD

Management

- Euvolaemia, nimodipine; **induced hypertension** if DCI
- Endovascular: intra-arterial vasodilators, angioplasty; avoid prophylactic HHH therapy

Hyponatraemia

- SIADH (euvolaemic) vs **cerebral salt wasting** (hypovolaemic, high urine Na + volume)
- Do not fluid restrict in SAH; saline/hypertonic, fludrocortisone for CSW

Key point: Never fluid restrict in SAH: hypovolaemia worsens DCI.

Notes: SAH p.1

Status epilepticus: definitions and treatment

ILAE definition

- Convulsive: **t1 = 5 min** (treat), t2 = 30 min (damage)
- Focal impaired awareness: t1 10 min, t2 60 min

Time-based management

- 0–5 min: ABC, glucose, thiamine
- 5–20: **benzodiazepine** (lorazepam 4 mg IV, midazolam 10 mg IM/buccal), repeat once
- 20–40: levetiracetam 60 mg/kg, valproate 40 mg/kg or fosphenytoin 20 mg PE/kg (ESETT: equal)
- > 40: anaesthetic (propofol, midazolam, thiopental) + EEG

Systemic complications

- Hypoxia, aspiration, rhabdomyolysis, lactic acidosis, hyperthermia, arrhythmia, neurogenic oedema

Key point: Give an adequate benzodiazepine dose early; delay makes it refractory.

Refractory, super-refractory SE and NCSE

Definitions

- Refractory: fails benzodiazepine + second-line; super-refractory: ≥ 24 h despite anaesthesia
- **NORSE**: new-onset refractory SE without cause (autoimmune, FIRES)

Super-refractory management

- Continuous EEG; anaesthetic to burst suppression 24–48 h, slow wean
- Ketamine, add other ASMs, steroids/IVIG/PLEX if autoimmune, ketogenic diet

NCSE

- Unexplained coma or fluctuating consciousness, subtle twitching
- **EEG required** (Salzburg criteria); treat like SE

Key point: Unexplained coma after seizures → EEG for NCSE.

Intracerebral haemorrhage

ICH score (0–6)

- GCS 3–4 (2), 5–12 (1); volume ≥ 30 mL (1); IVH (1); infratentorial (1); age ≥ 80 (1)
- Score 3 ≈ 72% 30-day mortality; don't use alone to limit care

Acute management

- **SBP 140 (INTERACT2/3)**, avoid falls > 70 mmHg; reverse anticoagulation (PCC, idarucizumab, andexanet)
- Glucose, temperature, seizures if clinical, VTE prevention with IPC

Surgery

- Cerebellar > 3 cm or brainstem compression/hydrocephalus: evacuate
- Lobar superficial clots: MIS (ENRICH); EVD for hydrocephalus

Key point: Bundle care (BP, glucose, temperature, reversal) improves ICH outcome.

Notes: ICH p.4

TBI: secondary injury and BTF targets

Injury types

- Primary: at impact; secondary: hypoxia, hypotension, raised ICP, hyper/hypocapnia, fever, seizures, glucose

Immediate management

- Intubate if GCS ≤ 8; SpO2 > 94%, PaCO2 4.5–5 kPa
- **SBP ≥ 100 (50–69 yr) or ≥ 110 (15–49, > 70)**; CT; neurosurgery

BTF targets

- ICP < 22 mmHg, CPP 60–70, Na 140–145, glucose 6–10, temp < 38
- Seizure prophylaxis 7 days (levetiracetam/phenytoin)

Key point: One episode of hypotension or hypoxia doubles TBI mortality.

Raised ICP: Monro–Kellie, SIBICC and osmotherapy

Monro–Kellie

- Fixed skull: brain + blood + CSF; compensation then steep rise on pressure–volume curve

SIBICC tiers

- Tier 0–1: head up, sedation, normocapnia, EVD drainage, osmotherapy
- Tier 2: mild hypocapnia (4.3–4.6 kPa), NMB, MAP challenge
- Tier 3: barbiturate coma, decompressive craniectomy (RESCUEicp), hypothermia

Mannitol vs hypertonic saline

- Mannitol 0.25–1 g/kg: osmotic diuresis, hypotension; check osmolar gap
- HTS 3%/23.4%: volume expansion, no diuresis, hypernatraemia risk

Key point: Escalate in tiers; HTS preferred if hypovolaemic.

Ischaemic stroke: thrombolysis and thrombectomy

IV thrombolysis

- **Within 4.5 h** (or wake-up with DWI–FLAIR mismatch); tenecteplase 0.25 mg/kg or alteplase 0.9 mg/kg
- BP < 185/110 before, < 180/105 for 24 h after
- Contraindications: haemorrhage, recent surgery/bleeding, platelets < 100, INR > 1.7

Thrombectomy

- LVO within 6 h; up to 24 h with mismatch (DAWN, DEFUSE-3); large core (SELECT2)

Post-reperfusion care

- Neuro checks, BP, glucose, temperature, swallow, haemorrhagic transformation (CT at 24 h)

Key point: Time is brain: door-to-needle < 60 min.

Malignant MCA infarction and TIA

Malignant MCA

- > 50% MCA territory, early deterioration from oedema (days 2–5)
- **Decompressive hemicraniectomy within 48 h** in ≤ 60 yr (DECIMAL, DESTINY, HAMLET); DESTINY II > 60

Medical measures

- Head up, normothermia, glucose, osmotherapy as bridge, avoid hypotonic fluids

TIA

- ABCD2 limits; urgent imaging, carotid Doppler, ECG monitoring
- DAPT (aspirin + clopidogrel 21 days) for minor stroke/TIA (CHANCE, POINT)

Key point: Early hemicraniectomy saves life in malignant MCA infarction.

Spinal cord injury and neurogenic shock

ASIA classification

- A complete; B sensory incomplete; C motor incomplete < 3; D motor ≥ 3; E normal

Neurogenic vs spinal shock

- Neurogenic: hypotension + bradycardia (loss of sympathetic tone, injury above T6)
- Spinal shock: flaccid areflexia temporarily below injury
- MAP 85–90 for 5–7 days; noradrenaline; atropine/pacing for bradycardia

Respiratory management

- C3–5: diaphragm; monitor VC (< 15 mL/kg → intubate); early tracheostomy

Key point: Lesions above C5 often need ventilation: monitor VC closely.

Guillain–Barré syndrome

Diagnosis

- Ascending flaccid weakness, areflexia; CSF high protein, normal cells
- Variants: AIDP, AMAN/AMSAN, Miller Fisher (ophthalmoplegia, ataxia, areflexia)

Intubation (20/30/40 rule)

- **VC < 20 mL/kg, MIP weaker than −30 cmH2O, MEP < 40**
- Bulbar dysfunction, rapid progression, autonomic instability

Treatment

- **IVIG 0.4 g/kg × 5 days** or plasma exchange (equal); no steroids
- VTE prophylaxis, pain, autonomic monitoring, physio

Key point: Measure VC serially: don't wait for hypoxia.

Notes: GBS p.4

Myasthenic and cholinergic crisis

Myasthenic crisis

- Respiratory failure from MG weakness; triggers: infection, surgery, drugs (aminoglycosides, Mg)

Differentiate

- Cholinergic: excess pyridostigmine, miosis, salivation, bradycardia, fasciculations
- Myasthenic: no muscarinic signs; ice pack test; EMG

Management

- Intubate if VC < 20 mL/kg; stop pyridostigmine while ventilated
- **IVIG or plasma exchange**, then steroids/immunosuppression

Key point: Avoid aminoglycosides and NMB drugs in MG.

Duchenne muscular dystrophy in ICU

Respiratory and cardiac

- Restrictive lung disease, weak cough, sleep hypoventilation; FVC falls
- Dilated cardiomyopathy, arrhythmias

ICU respiratory care

- NIV early, mechanical insufflation–exsufflation (cough assist), extubate to NIV

Anaesthetic concerns

- **Avoid suxamethonium (hyperkalaemia, rhabdo)** and volatile agents
- TIVA, careful NMB use, cardiac assessment

Key point: Suxamethonium can kill DMD patients: never use it.

Paroxysmal sympathetic hyperactivity

Definition

- Episodic tachycardia, hypertension, fever, sweating, posturing after brain injury (TBI)
- Disconnection of inhibitory pathways → excitatory reflexes

Diagnosis

- **PSH-AM score** (clinical features + diagnosis likelihood)
- Exclude sepsis, withdrawal, seizures

Management

- Avoid triggers; opioids, gabapentin, propranolol, clonidine
- Bromocriptine, intrathecal baclofen

Key point: PSH is a diagnosis of exclusion: rule out sepsis first.

Bacterial meningitis

First hour

- Blood cultures, **dexamethasone 10 mg 6-hourly** (pneumococcus), ceftriaxone 2 g 12-hourly
- Add ampicillin (> 50 yr, Listeria), vancomycin if resistance; acyclovir if encephalitis

CT before LP

- Focal signs, GCS < 10, seizures, papilloedema, immunocompromise

CSF comparison

- Bacterial: high neutrophils, protein > 1 g/L, glucose < 40%
- Viral: lymphocytes, normal glucose; TB: lymphocytes, very high protein, low glucose

Key point: Never delay antibiotics for CT or LP.

Notes: meningitis ,aseptic meningitis ,ventriculitis p.4

Healthcare-associated ventriculitis

Definition

- Infection of ventricles after EVD/surgery: CSF pleocytosis, low glucose, positive culture
- Risks: EVD > 5 days, frequent sampling, CSF leak, IVH

Management

- **Vancomycin + meropenem/cefepime**; intraventricular antibiotics if refractory
- Remove/replace device

EVD prevention

- Aseptic insertion, tunnelling, antibiotic-impregnated catheters, limit sampling, closed system

Key point: Limit EVD sampling and duration to prevent ventriculitis.

Notes: meningitis ,aseptic meningitis ,ventriculitis p.5

HSV encephalitis

Diagnosis

- Fever, confusion, seizures, focal signs
- **CSF HSV PCR**, MRI temporal lobe changes, EEG periodic discharges

Treatment

- **IV acyclovir 10 mg/kg 8-hourly for 14–21 days**; start empirically
- Adjust for renal function; repeat PCR

Complications

- Seizures, memory loss, autoimmune NMDA encephalitis later; mortality ~ 70% untreated

Key point: Start acyclovir empirically in encephalitis: don't wait.

ICU delirium and the ABCDEF bundle

Diagnosis

- **CAM-ICU**: acute onset/fluctuation + inattention + (altered LOC or disorganised thinking)
- ICDSC ≥ 4

Risk factors and consequences

- Age, dementia, sepsis, benzodiazepines, sleep loss
- ↑ Mortality, long-term cognitive impairment, longer stay

ABCDEF bundle

- Assess pain, SAT + SBT, choice of sedation, delirium monitoring, early mobility, family
- Haloperidol not effective (AID-ICU), dexmedetomidine for agitation

Key point: Prevent delirium: light sedation, mobility, sleep, family.

Notes: delirium and sedation p.1

Analgesia-first sedation and PRIS

Principles

- Analgesia first, light sedation (RASS 0 to −2), daily awakening

Drugs

- Propofol: rapid offset, hypotension
- Dexmedetomidine: rousable, less delirium, bradycardia (MENDS2, SPICE III)
- Benzodiazepines: more delirium, accumulation → avoid

Propofol infusion syndrome

- > 4 mg/kg/h > 48 h; metabolic acidosis, rhabdo, hyperkalaemia, arrhythmia, AKI
- Stop propofol, supportive, RRT, ECMO

Key point: Avoid benzodiazepines; watch propofol dose and lactate.

Brainstem death testing and donor management

Preconditions

- Known cause of irreversible brain damage; exclude sedation, hypothermia, metabolic/endocrine causes
- Temperature > 34 °C, MAP > 60, normal Na, glucose, $PaCO_2$

Clinical tests

- Absent pupil, corneal, oculovestibular, gag, cough reflexes; no motor response to pain
- **Apnoea test**: $PaCO_2$ rises > 6.65 kPa with pH < 7.4 and no breathing

Donor management

- MAP > 60, DI with DDAVP, hormone replacement, protective ventilation

Key point: Brain death needs strict preconditions before testing.

Guidelines (no 2026 note on this topic)